THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD



COURSE TITLE	Computer Networks
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LAB TITLE	Sequence Diagram
STUDENT NAME	MAHDI ALI NAHVI
STUDENT ROLL NUMBER	2024-SE-16
INSTRUCTOR NAME	Engr. Dr. Asma Javaid
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1. Objective

The objective of this lab is to model the dynamic behaviour of two systems using UML sequence diagrams. A sequence diagram shows how objects interact with each other over time, using messages arranged in the order in which they occur. The two systems modelled are:

- Q1: The data processing involved when a customer withdraws cash from an ATM.
- Q2: An online ticket reservation system.

Notation used

Element	Meaning
Actor / Object box	A participant in the interaction, shown at the top of its lifeline. The actor is the external user; objects are system components.
Lifeline (dashed vertical line)	The existence of a participant over time. Time flows from top to bottom.
Solid arrow	A synchronous call or request message sent from one participant to another.
Dashed arrow	A return message or reply to an earlier request.
alt frame	A combined fragment that shows alternative flows. Only one branch runs, depending on its guard condition.

2. Question 1: ATM Cash Withdrawal

Problem statement

Draw a sequence diagram that models the data processing involved when a customer withdraws cash from the ATM.

Participants

Participant	Role
Customer (actor)	The bank customer who inserts the card, enters the PIN and requests cash.
:ATM Interface	The machine's screen, keypad and card reader. It collects input and displays results.
:Bank Server	Authenticates the customer, checks balance and limits, and approves or declines the withdrawal.
:Account Database	Stores account data. It verifies credentials, returns the balance and records the debit.
:Cash Dispenser	The hardware unit that physically dispenses the notes.

Assumptions

- The card is valid and readable, and the ATM is online with enough cash loaded.
- The PIN is correct. Handling of wrong PIN attempts and card retention is outside the scope of this diagram.

- The bank checks the account balance and the daily withdrawal limit before approving.

Description of the flow

Authentication (messages 1 to 8). The customer inserts the card and the ATM asks for the PIN. The ATM sends the card number and PIN to the Bank Server, which checks them against the Account Database. After a successful reply the ATM shows the transaction menu.

Balance check (messages 9 to 12). The customer chooses Withdraw and enters an amount. The ATM sends a balance check request to the Bank Server, which reads the current balance from the Account Database.

Approved branch (messages 13 to 18). If the balance is at least the requested amount and the daily limit is not exceeded, the bank debits the account and the database confirms the update. The bank approves the withdrawal, the ATM instructs the Cash Dispenser to release the notes, and the customer receives the cash, a receipt and the returned card.

Declined branch (messages 19 and 20). If funds are insufficient or the limit is exceeded, the bank declines the request. No debit is made and no cash is dispensed. The ATM shows an error message and ejects the card.

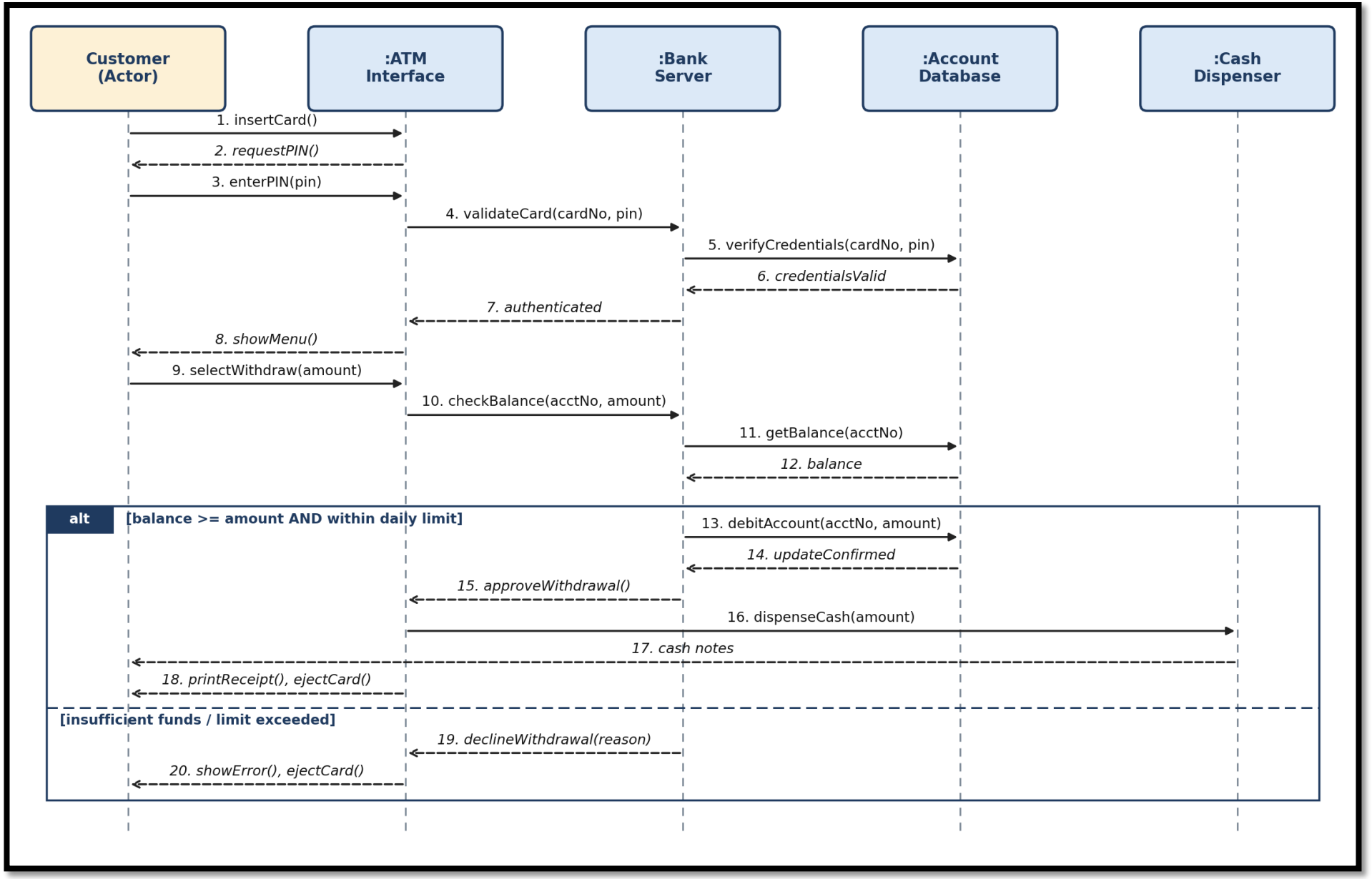


Figure 1: Sequence diagram of cash withdrawal from an ATM

3. Question 2: Online Ticket Reservation System

Problem statement

Draw a sequence diagram for the online ticket reservation system.

Participants

Participant	Role
Customer (actor)	The user who searches for trips, selects seats, enters details and pays.
:Web Interface	The website or app front end that takes user input and shows results.
:Reservation System	The core business logic. It coordinates availability, seat locking, payment and ticket issue.
:Ticket Database	Holds trips, seat inventory and bookings.
:Payment Gateway	The external service that authorizes the card payment.
:Notification Service	Sends the booking confirmation and e-ticket to the customer by email or SMS.

Assumptions

- The customer is already logged in or books as a guest.
- Selected seats are locked temporarily while payment is processed, so two customers cannot book the same seat.
- Payment is handled by an external gateway and the outcome is either approved or declined.

Description of the flow

Search (messages 1 to 6). The customer enters the route and date. The Web Interface passes the request to the Reservation System, which queries the Ticket Database for available seats. The trip list is returned and displayed.

Seat selection (messages 7 to 10). The customer picks a trip and seats. The Reservation System asks the database to lock those seats and receives confirmation.

Payment (messages 11 to 13). The customer enters passenger details and confirms. The Reservation System sends the amount and card details to the Payment Gateway for authorization.

Payment approved (messages 14 to 18). The booking is confirmed and the ticket saved in the database. The Notification Service sends a confirmation, and the customer sees the e-ticket on screen.

Payment declined (messages 19 to 22). The Reservation System releases the locked seats so others can book them, and the customer is shown an error with the option to retry.

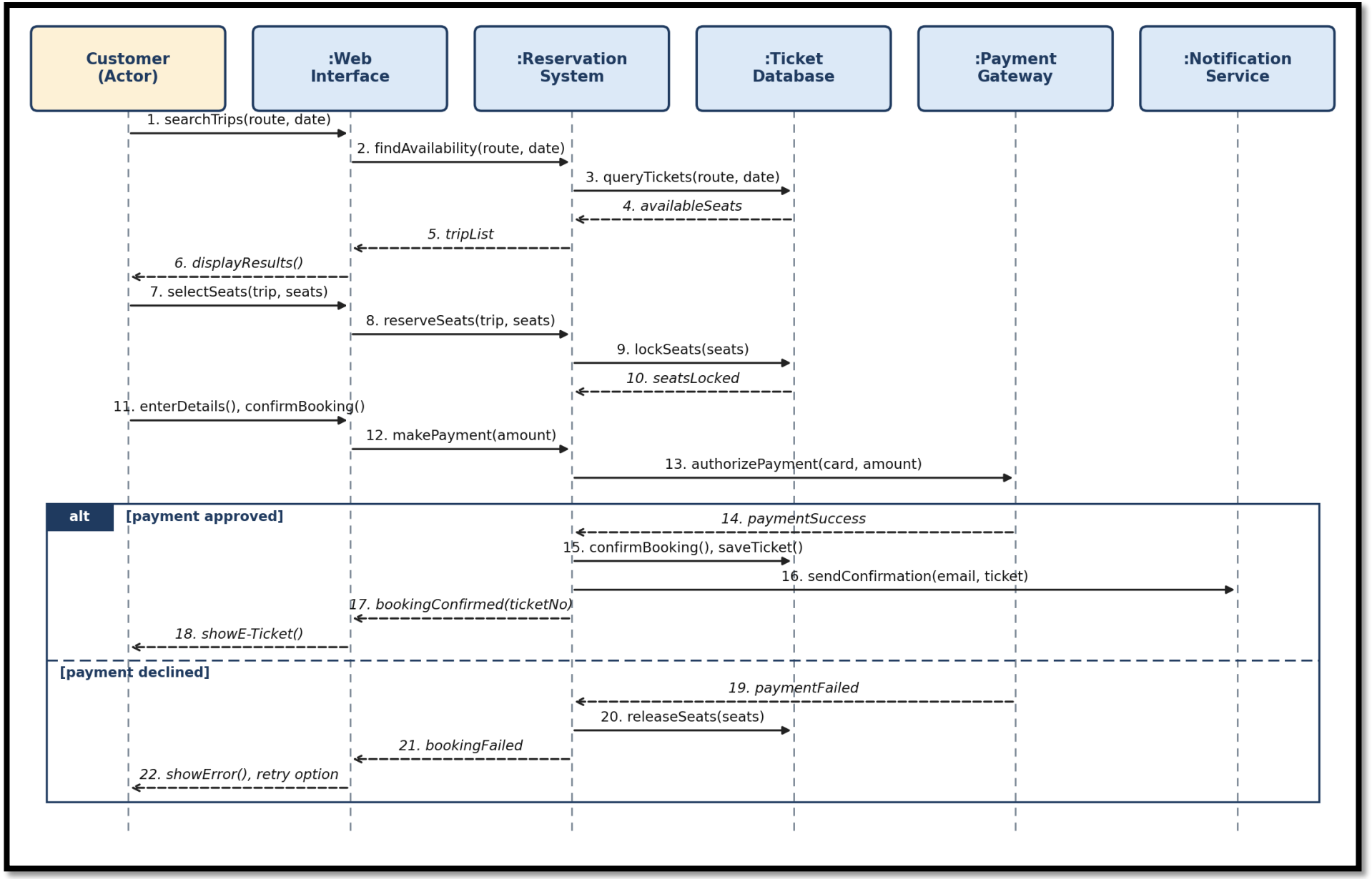


Figure 2: Sequence diagram of the online ticket reservation system

4. Conclusion

Sequence diagrams were drawn for both systems. The ATM diagram shows the ordered exchange between the customer, ATM, bank server, database and cash dispenser, including the alternative flow for insufficient funds. The ticket reservation diagram shows search, seat locking, payment and confirmation, with a separate branch that releases seats when payment fails.

In both cases the alt fragment captures the success and failure paths. Modelling these interactions before coding makes responsibilities clear and exposes missing steps early, which supports the construction phase of software development.